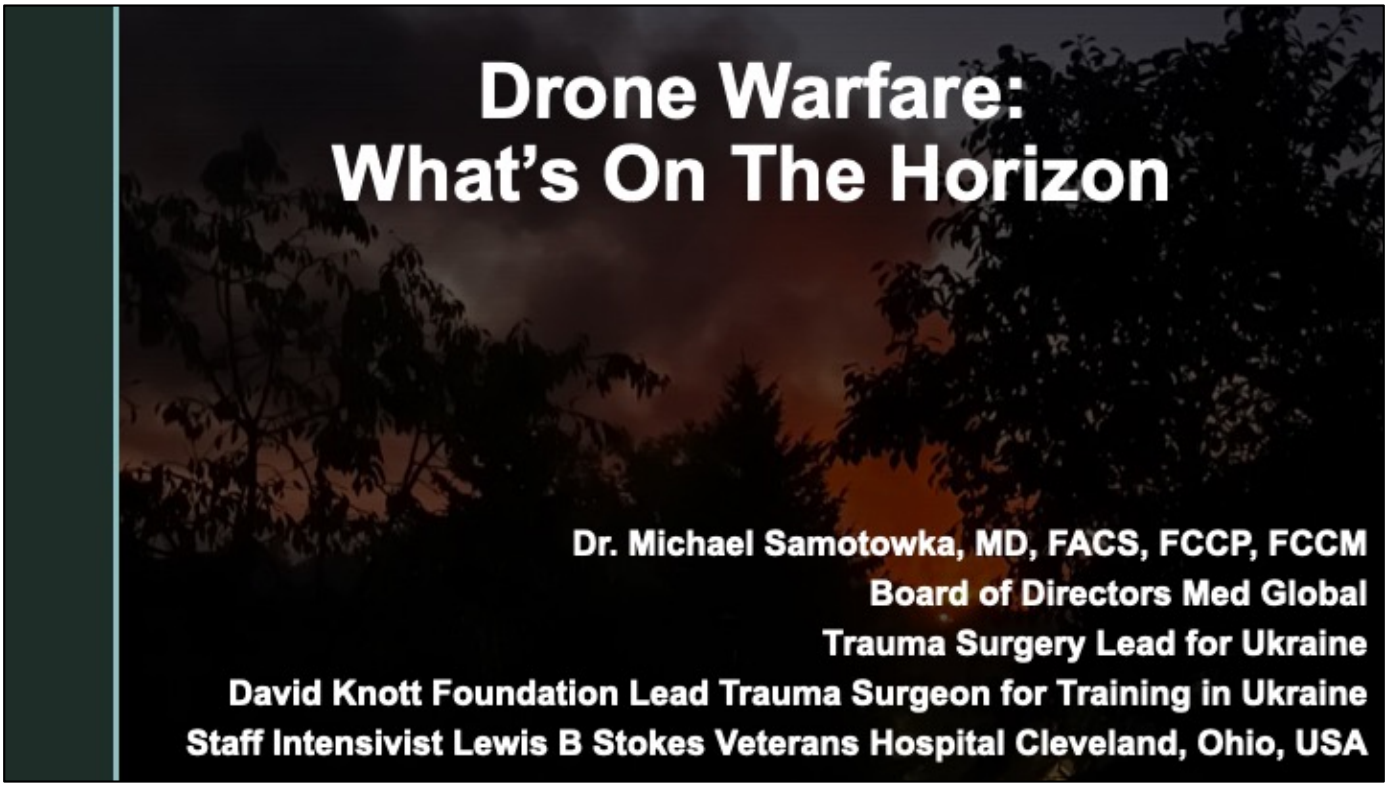
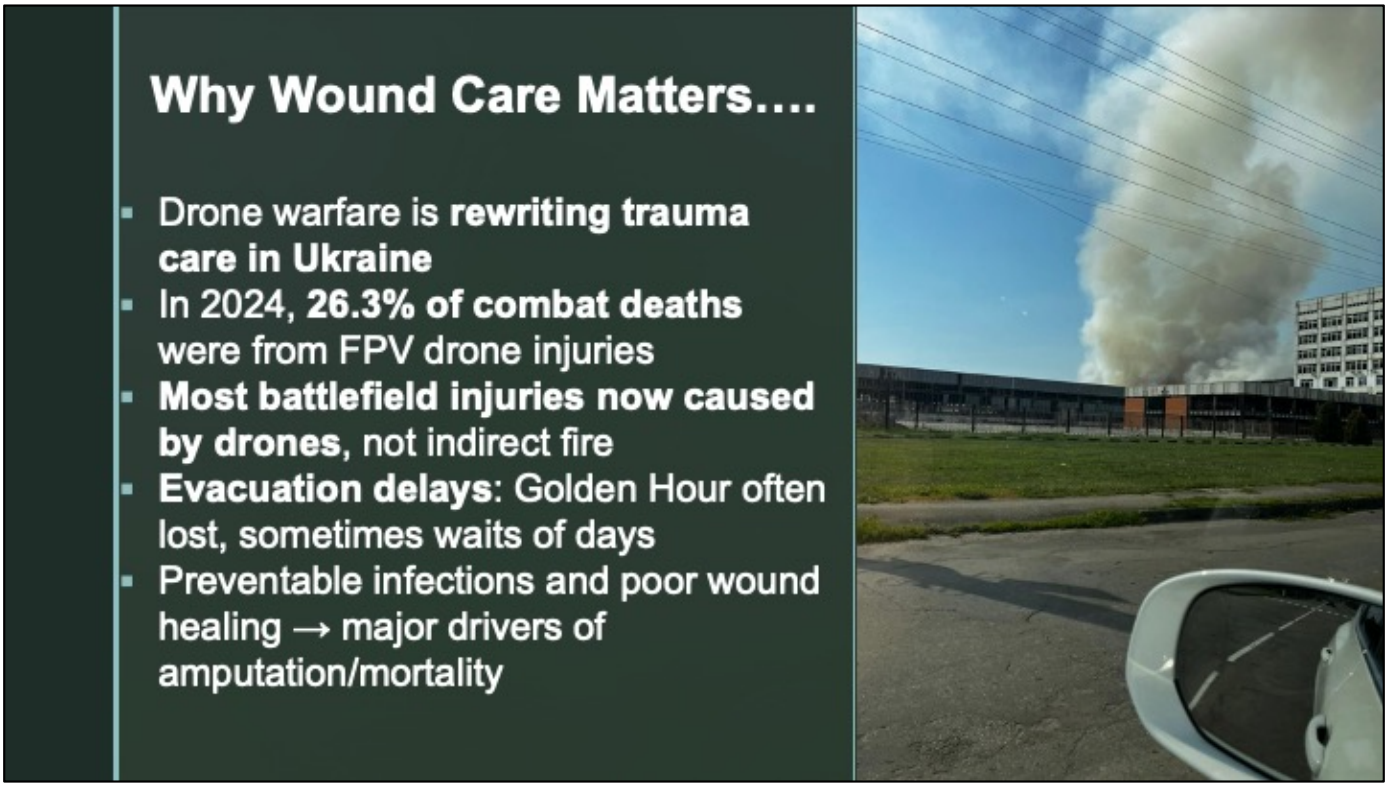
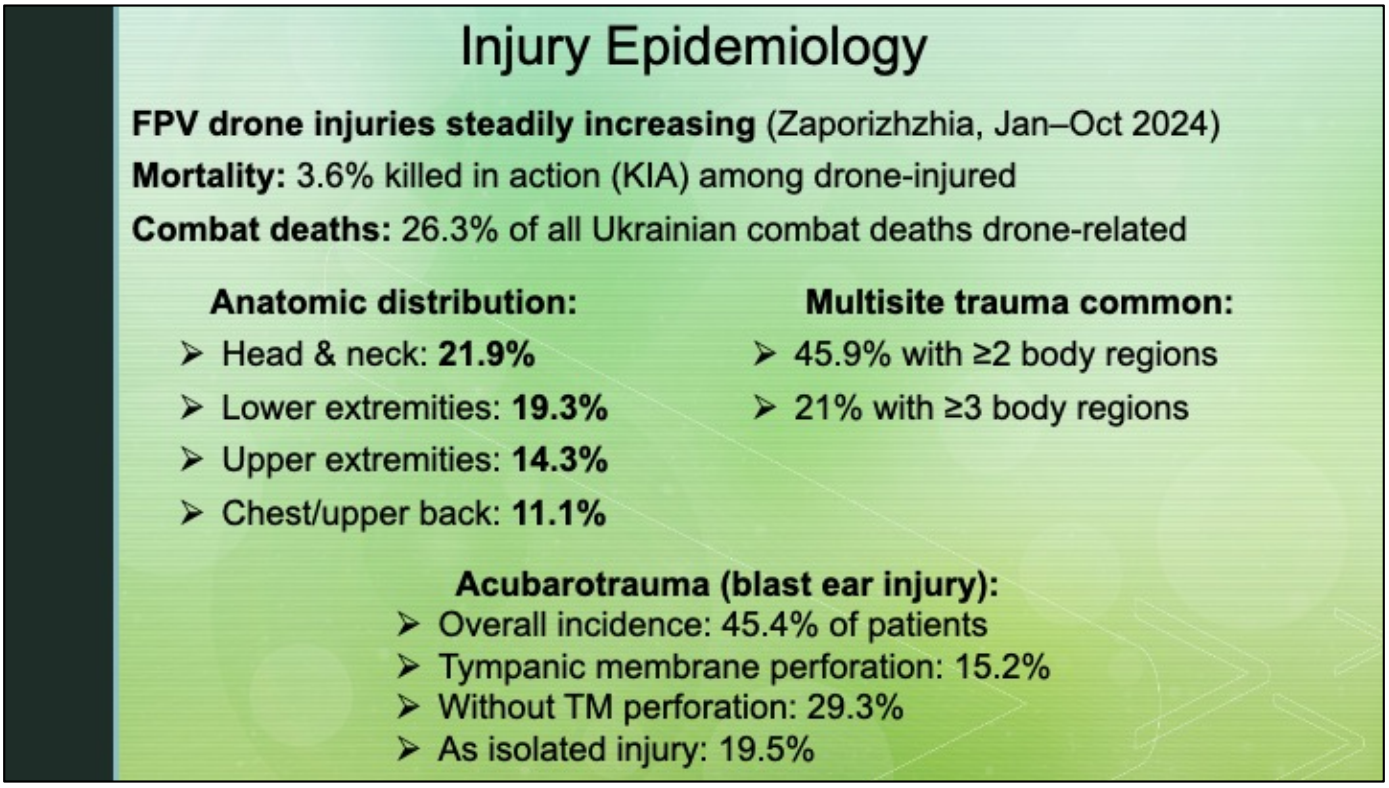


- Today we're not just talking about wound care — we're talking about how **drone warfare is rewriting medicine itself**. Ukraine is the first place in history where drones have dominated the battlefield on an industrial scale, and it's changing everything: the injuries we see, how patients arrive, and even how hospitals must be built and protected. This presentation will focus on **hospital-based wound care strategies** for these injuries — practical lessons learned in Ukraine — but it also carries a bigger message: these lessons are directly relevant to **NATO readiness**.
  Because what Ukraine faces today, NATO forces may face tomorrow.


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- Drone warfare has become the defining cause of trauma in Ukraine. By 2024, more than a **quarter of all combat deaths — 26.3% — were due to FPV drone injuries**.
- Almost all battlefield injuries now come from drones, not artillery or indirect fire.
- These injuries also arrive **late**. Medics often cannot evacuate for hours or even days, which means the Golden Hour is frequently lost.
- That delay dramatically increases the risk of infection, sepsis, and amputations — preventable causes of death if wound care is aggressive and systematic.



Hard data shows how common these injuries are. From Zaporizhzhia, among drone casualties: **3.6% died on scene**, and drones accounted for **over 26% of all combat deaths**. In terms of anatomic distribution:

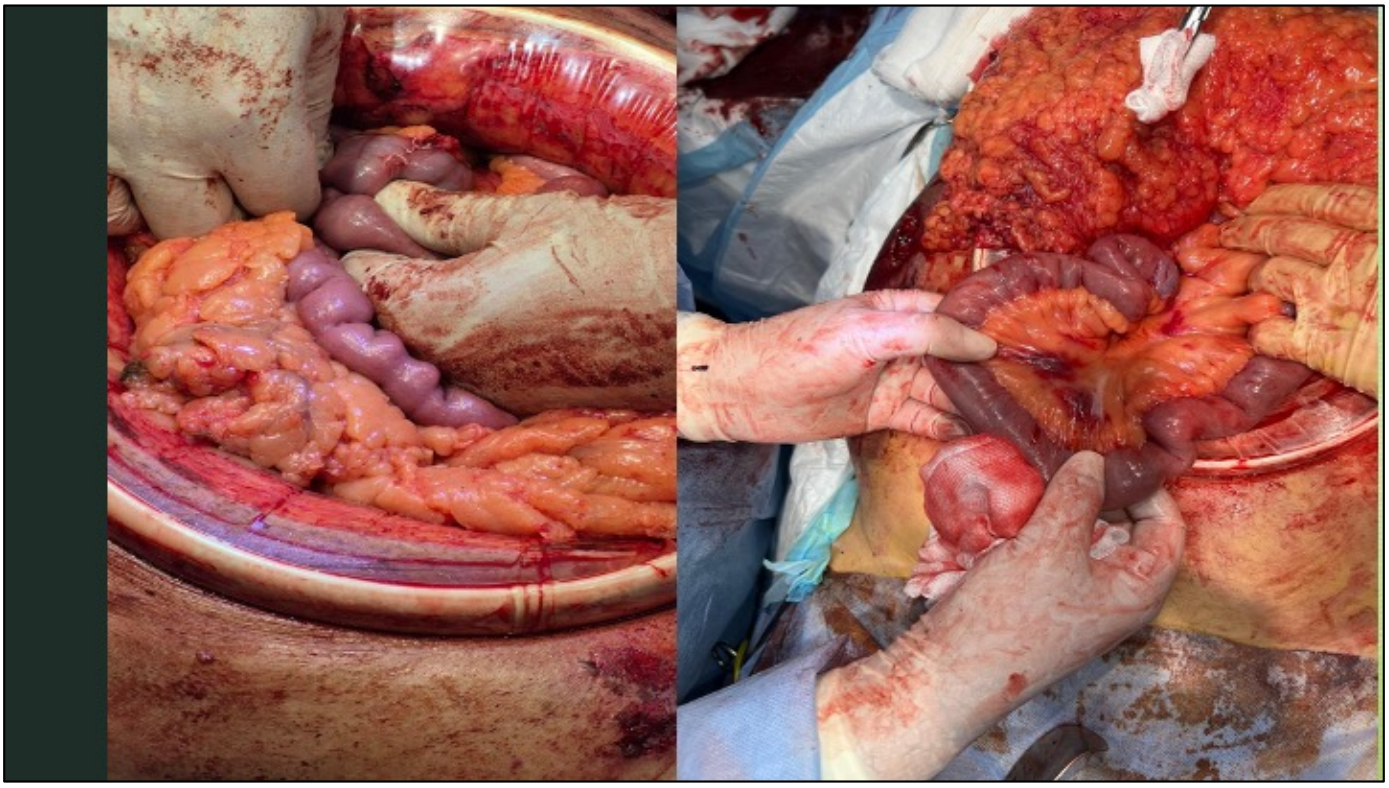
- **Head and neck injuries lead at 21.9%** — and these include ear and facial trauma.
- **Lower extremities: 19.3%**, **upper extremities: 14.3%**, **chest/back: 11.1%**.
- Nearly half — **45.9% — had multiple regions injured**, and **1 in 5 had three or more**.
- And here's the key point on **ear injuries**: almost **half of all patients (45.4%) had acubarotrauma — primary blast injury to the ear**.
- About **15% had tympanic membrane perforations**,
- Another **29% had inner-ear damage without perforation**,
- **1 in 5 had ear injury as their only injury**.
  This is huge. Ear injuries are not minor — they cause hearing loss, vertigo, disorientation, and can impact long-term rehabilitation.




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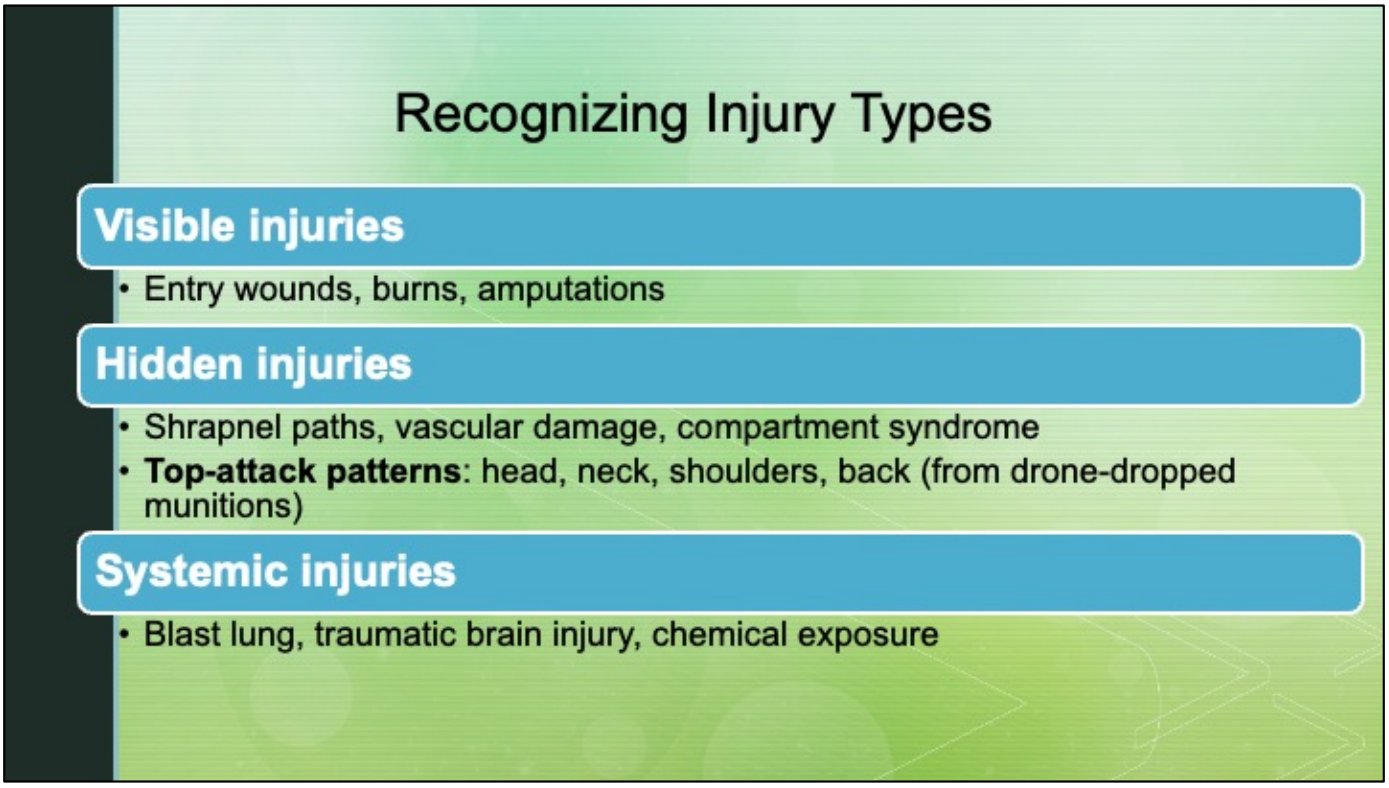




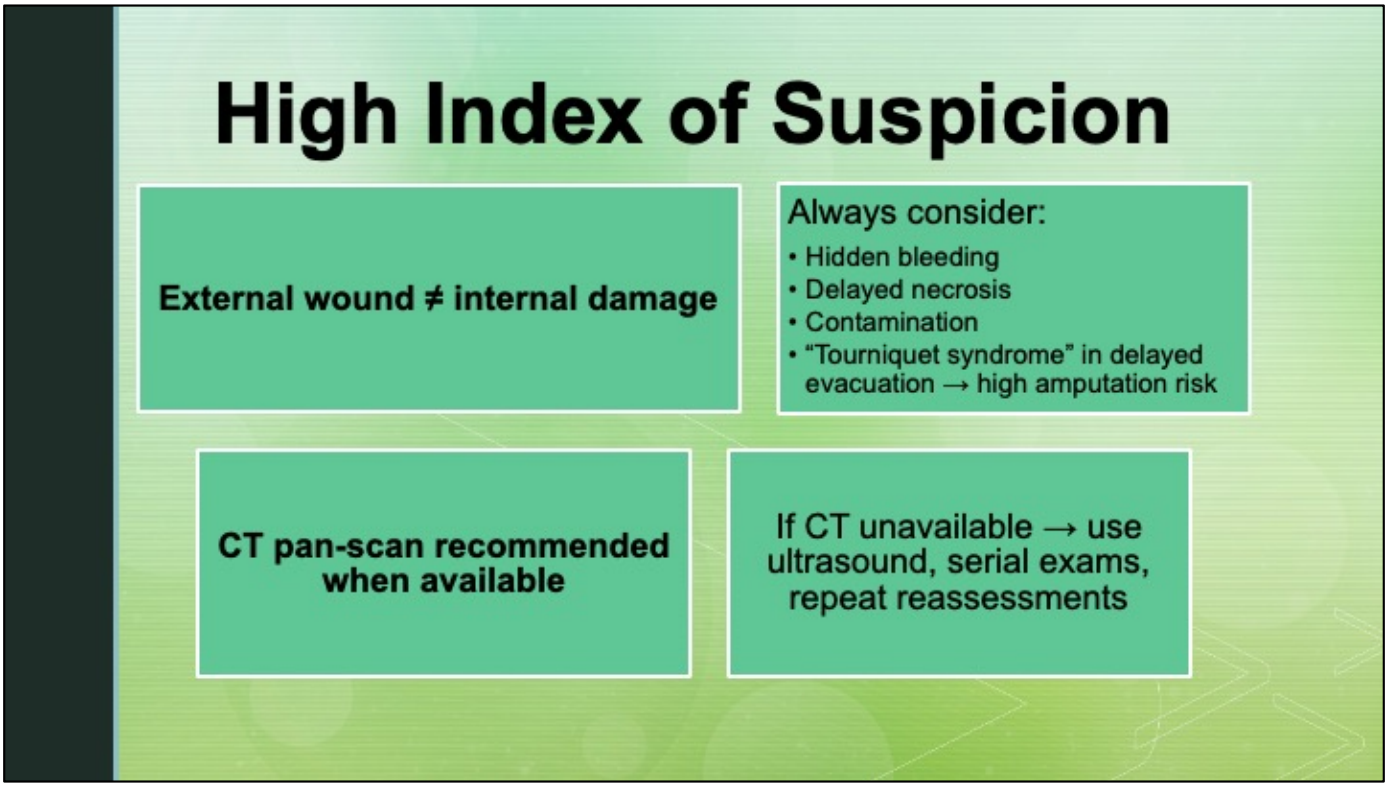




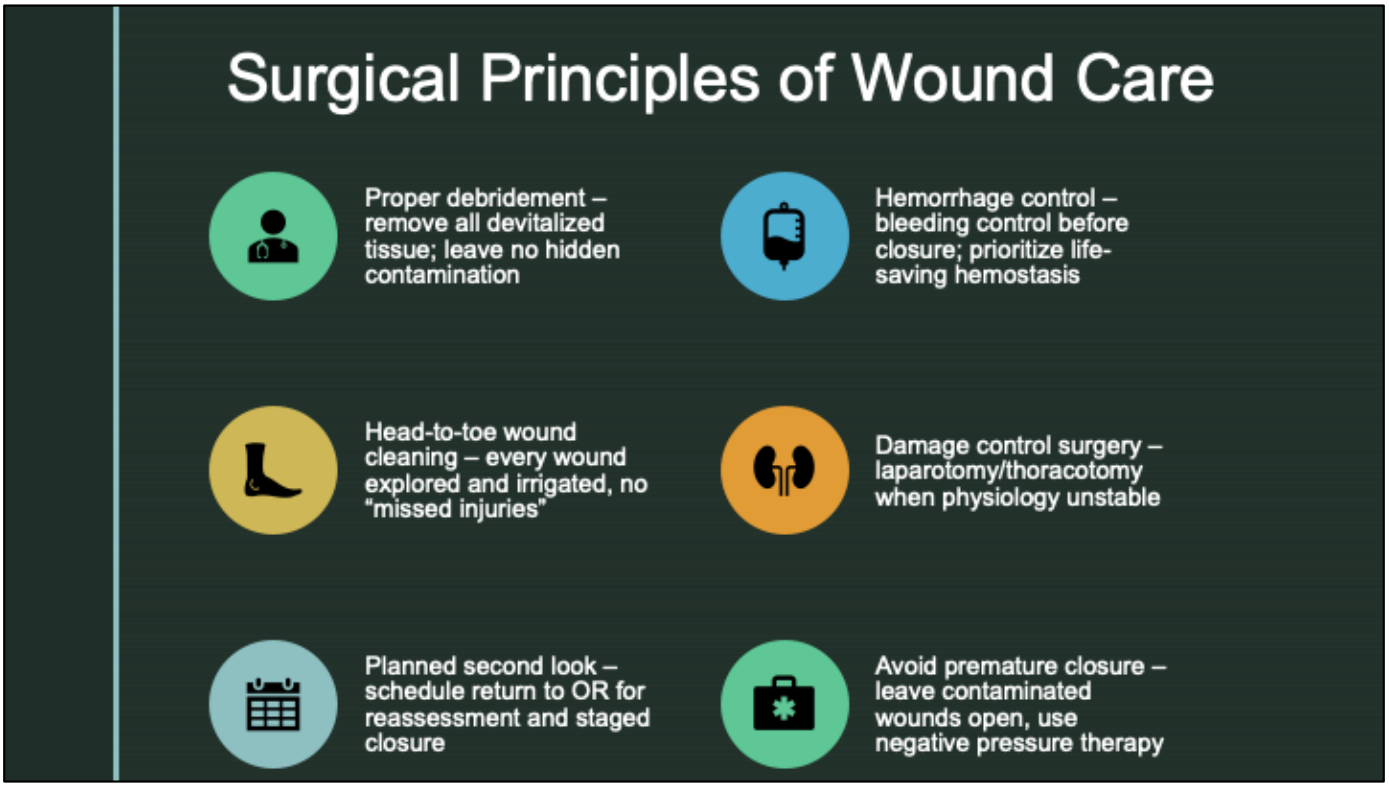
- Traditional body armor was designed for bullets and frontal fire. It protects the **chest and back**, but drones don't attack like that.
- **FPV drones detonate above soldiers.** The blast shatters and fragments rain downward, bypassing plates and striking into the **armpits, groin, head, neck, and ears**.
- These areas contain critical vessels and structures — the **axillary and femoral arteries, the brachial plexus, the carotid arteries, the airway, and the ears**. Injuries here are devastating and often fatal.
- Some Ukrainian units are adapting by using enhanced protective gear like **'Rarog Armor'**, modeled after Samurai armor. It extends coverage to the armpits, groin, and neck — the very gaps drones exploit.
- It costs around **$1,200 to $2,500 per set**, but in practice, clinicians are reporting it **saves lives** by preventing catastrophic bleeding from otherwise unprotected zones.



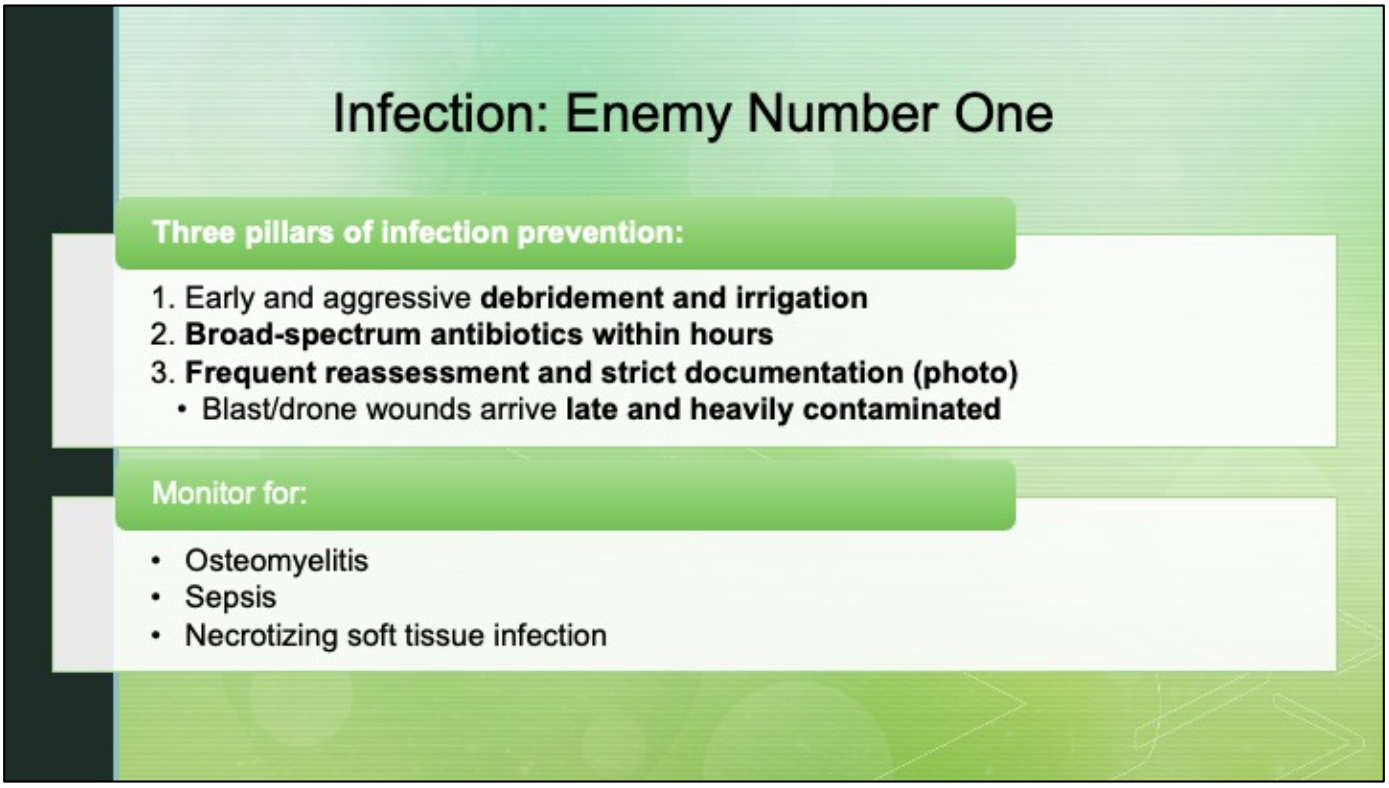
Drone munitions often strike from above, leading to **top-attack patterns** — injuries to the head, neck, shoulders, and back. These come on top of the visible wounds — amputations, burns, and entry sites — and the hidden injuries like shrapnel tracks, vascular lacerations, and compartment syndrome. Finally, systemic injuries like blast lung and traumatic brain injury are also common, and must be anticipated even if external wounds look small.



These wounds demand a **high index of suspicion**. Not only do small wounds hide major internal injuries, but with **prolonged evacuation times**, we are now seeing **'tourniquet syndrome'** — where extended tourniquet use contributes to high rates of limb loss once patients finally reach surgical care. The best diagnostic tool is a **CT pan-scan** to catch hidden shrapnel tracks and vascular injuries. If CT is not available, rely on ultrasound and repeat physical exams.
The bottom line: *never trust the outside appearance of a blast or drone wound*.



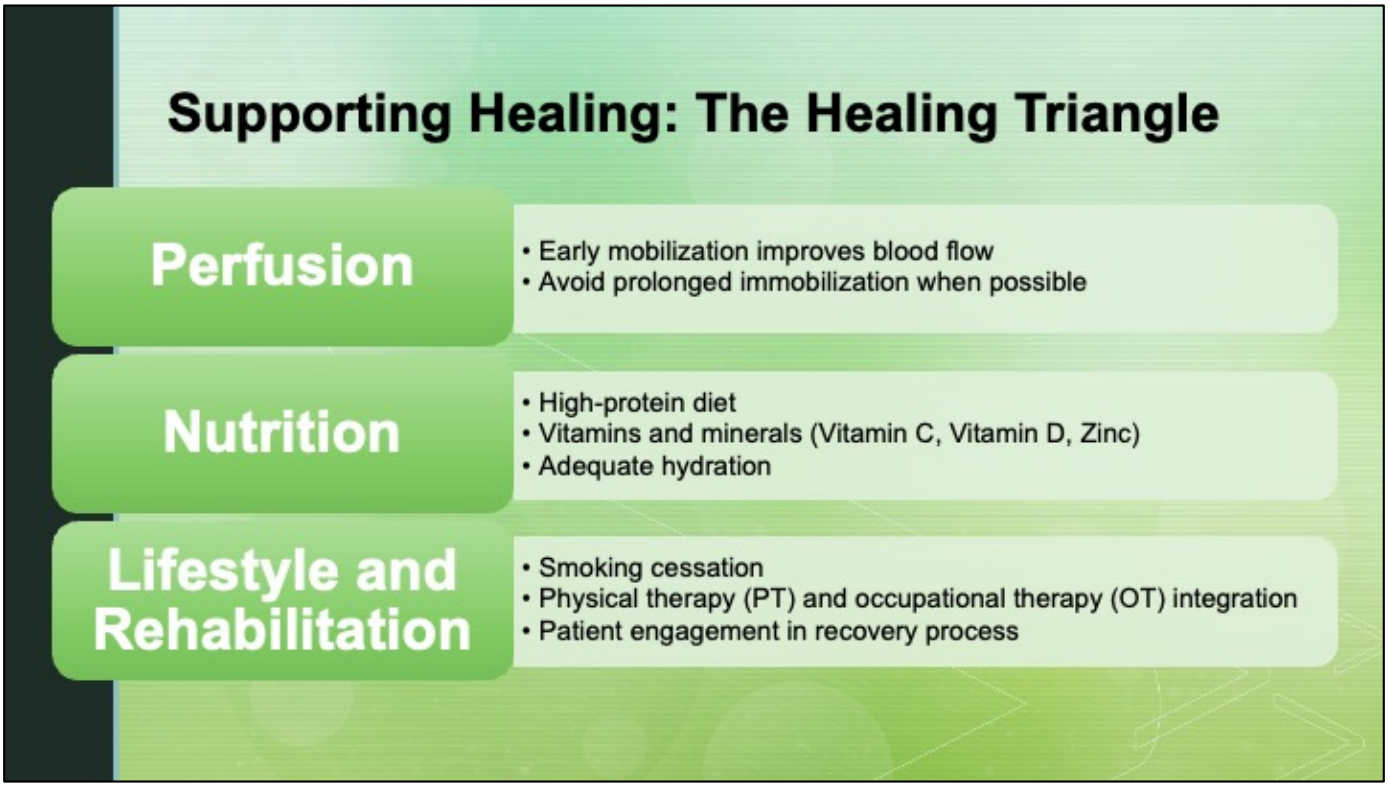
- This is where surgical discipline makes the difference.
- **First, proper debridement.** In drone injuries, tissue often looks viable on the surface but is already devitalized deeper. If you don't cut back until you see healthy bleeding tissue, infection is inevitable.
- **Second, hemorrhage control.** Bleeding is the most immediate killer — every wound must be inspected and controlled before closure.
- **Third, a proper head-to-toe exam.** These patients often have **multiple wounds** — remember the Zaporizhzhia data showed **45.9% with at least two regions involved**. If you only treat what you see, you'll miss life-threatening injuries.
- **Fourth, damage control surgery.** Sometimes the patient is too unstable for definitive repair. That's when a quick **laparotomy or thoracotomy** to stop bleeding and contamination is the right move.
- And **fifth, always plan a second look.** These wounds evolve, and contamination reveals itself later. A staged approach with a planned return to the OR saves lives and limbs.
- The mistake is to rush closure. **Drone wounds should almost never be closed primarily** — leave them open, use negative pressure, and come back when you're sure.



Infection risk is even higher in drone and explosive injuries, because evacuation is often **delayed well beyond the Golden Hour**. That means wounds arrive already contaminated, sometimes with days of debris inside.

First, early and aggressive debridement and irrigation. Any devitalized tissue must be removed quickly, and wounds thoroughly irrigated. **Second, broad-spectrum antibiotics.** These should be started within hours of injury, not delayed until infection has already developed. **Third, frequent reassessment and documentation.** Wounds must be checked daily — sometimes more often — and changes carefully documented, because what looks clean today may deteriorate tomorrow.

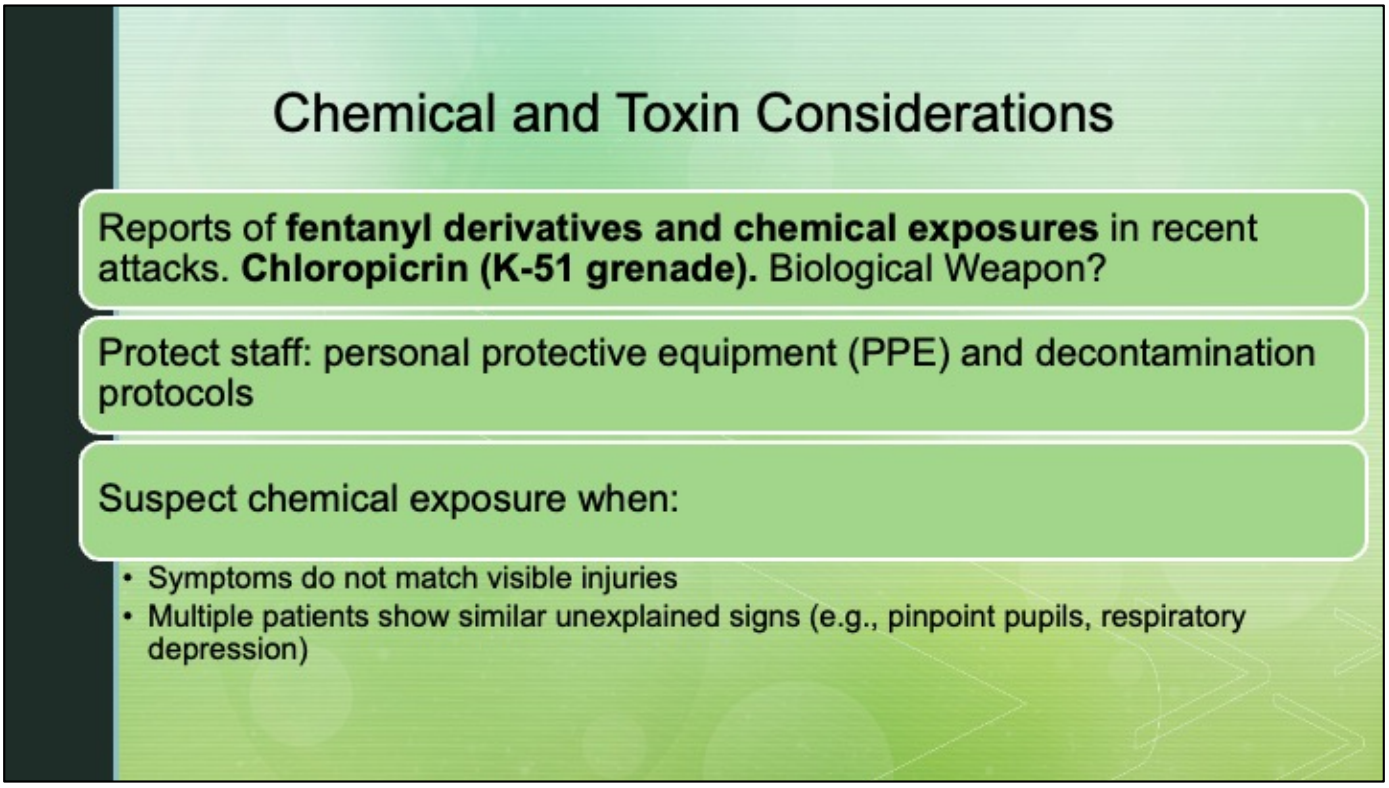
We must also watch closely for complications like **osteomyelitis, sepsis, and necrotizing infections**. Missed infection today means amputation or death tomorrow.



Successful wound healing is not just about surgery and dressings. It depends on the **healing triangle**: perfusion, nutrition, and lifestyle/rehabilitation.

First, **perfusion**. Early mobilization is one of the best ways to improve blood flow to injured tissue. Whenever possible, avoid prolonged immobilization. Encourage safe movement as soon as clinically appropriate. Second, **nutrition**. The body cannot heal without protein, vitamins, and minerals. Trauma patients often need higher protein intake, as well as Vitamin C, Vitamin D, and zinc. Don’t overlook hydration, which is critical for tissue recovery. Third, **lifestyle and rehabilitation**. Smoking greatly delays healing, so cessation support is vital. Physical therapy and occupational therapy should be integrated early to maintain function and promote recovery. Patients should also be engaged in their own recovery process — this improves adherence and outcomes.

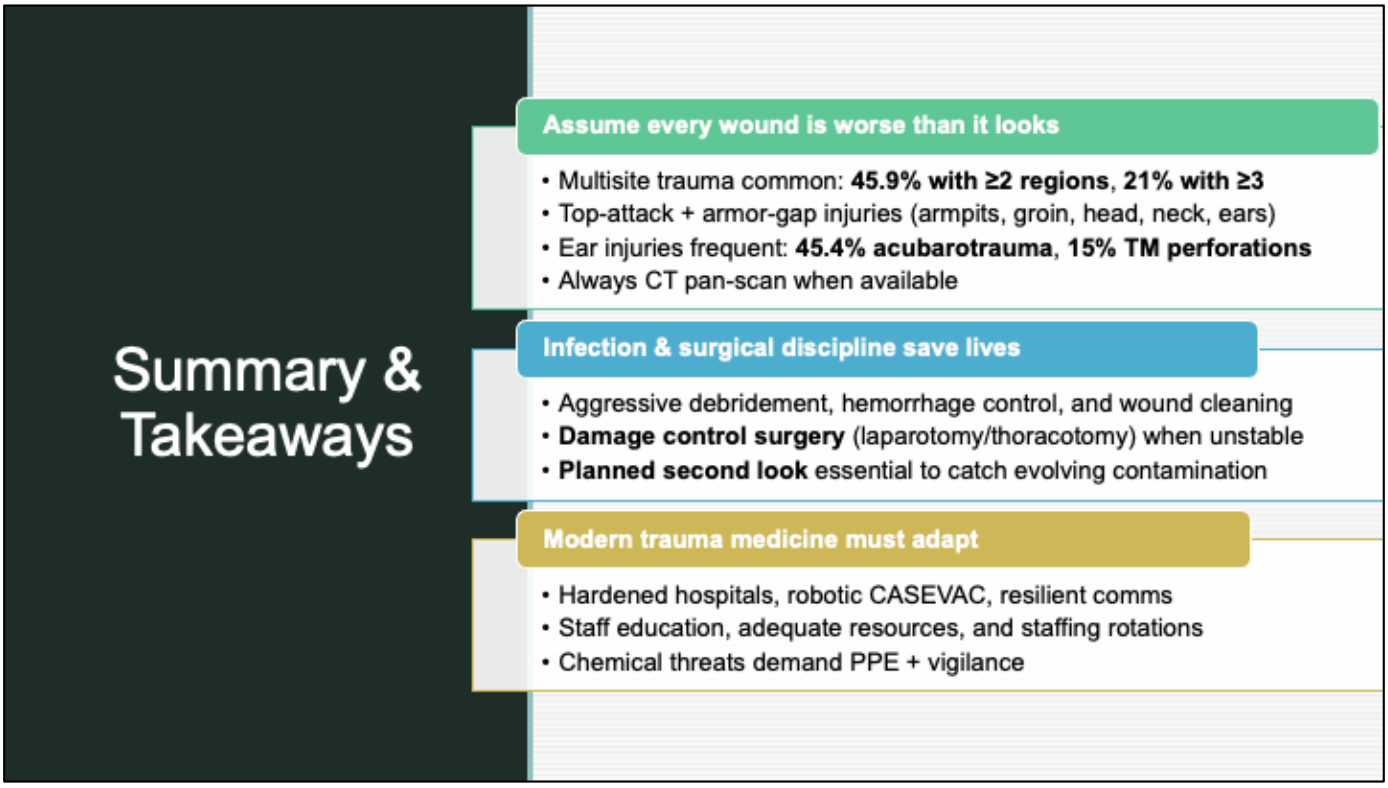
By addressing all three parts of the healing triangle, we give wounds the best chance to heal fully and quickly.



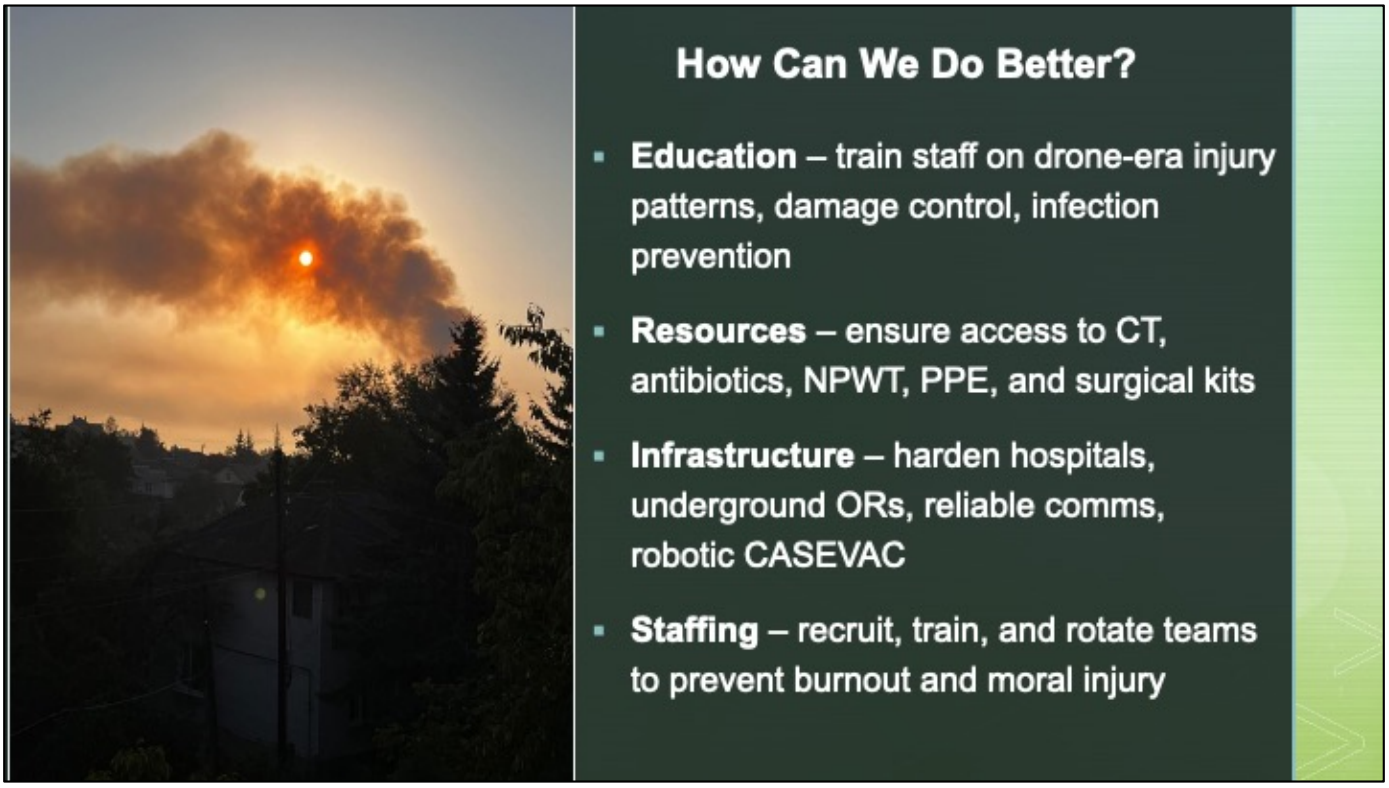
- In some cases, drone and explosive attacks have been linked with **chemical or toxin exposures**, including **fentanyl-like agents**. This means we not only need to care for the patient, but also to **protect the staff**. Always consider the use of **personal protective equipment and decontamination protocols** if there's a possibility of chemical contamination.

  Chloropicrin burns the airway and lungs. These patients bleed more during intubation, they desaturate quickly, and they don't ventilate well. Their wounds also heal slower, and their infection rates are higher — so what looks like 'just lung irritation' becomes a whole-body surgical problem. So when you see clustered patients with cough, hypoxia, or confusion, think toxic exposure — even if the wounds don't match.

Another red flag is when **multiple patients arrive with similar unexplained symptoms** that don't align with their injuries — this should raise suspicion for a chemical agent. Preparedness here is vital. We must balance treating the patient while ensuring the safety of the healthcare team.



- To close, let me leave you with three takeaways.
- **First, assume every wound is worse than it looks.** Almost half of drone casualties have multiple regions injured, and one in five have three or more. Injuries bypass armor — striking the armpits, groin, head, neck, and ears. Ear injuries alone are seen in nearly **half of patients**, with **15% suffering tympanic membrane rupture**. That's why CT pan-scan and repeated reassessment are essential.
- **Second, infection and surgical discipline save lives.** Proper debridement, hemorrhage control, and head-to-toe wound cleaning make the difference between life and limb loss. For unstable patients, **damage control surgery** — laparotomy or thoracotomy — comes first, with a **planned second look** to finish the job once physiology improves.
- **Third, we must adapt beyond the OR.** This means hardening hospitals, using robotic evacuation, and building resilient communications networks. It also means training and rotating staff, securing resources, and preparing for chemical threats like chloropicrin, which already account for **over 3% of exposures**.
- If we do all this, Ukraine can save more lives today — and NATO can be ready for the battles of tomorrow.



- “Up to this point, we’ve focused on the clinical and surgical details — but we have to ask: **how can we do better?**
- It starts with **education.** Everyone, from medics to surgeons, must be trained in **drone-era injury recognition, debridement, hemorrhage control, and staged operations**.
- Next is **resources.** Without antibiotics, negative pressure wound therapy, protective equipment, and reliable CT scanning, even the best-trained team can’t succeed.
- Then, **infrastructure.** We need hardened facilities, underground operating rooms, resilient communications, and robotic evacuation systems to keep patients and staff alive.
- And finally, **staffing.** This war creates nonstop mass-casualty cycles. Teams must be adequately manned, rotated, and supported to prevent burnout and moral injury.
- If we don’t address these areas, the best surgical techniques won’t matter. But if we do, then Ukraine — and NATO — can truly be ready for drone warfare.




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Is NATO Ready to face drone warfare?


GLORY TO UKRAINE
GLORY TO HEROES